SmartToken

AI-Assisted, Faculty-Controlled Participation and Token Management.

A technical system for structured participation, response evaluation and auditable token allocation.

React • Node.js • Express • PostgreSQL • JWT • Gemini • MediaRecorder

Problem → System Objective

CURRENT CHALLENGE

Classroom interruption

Faculty repeatedly handles participation requests manually.

Unstructured response records

Verbal participation is difficult to capture and audit.

Risk of automated decisions

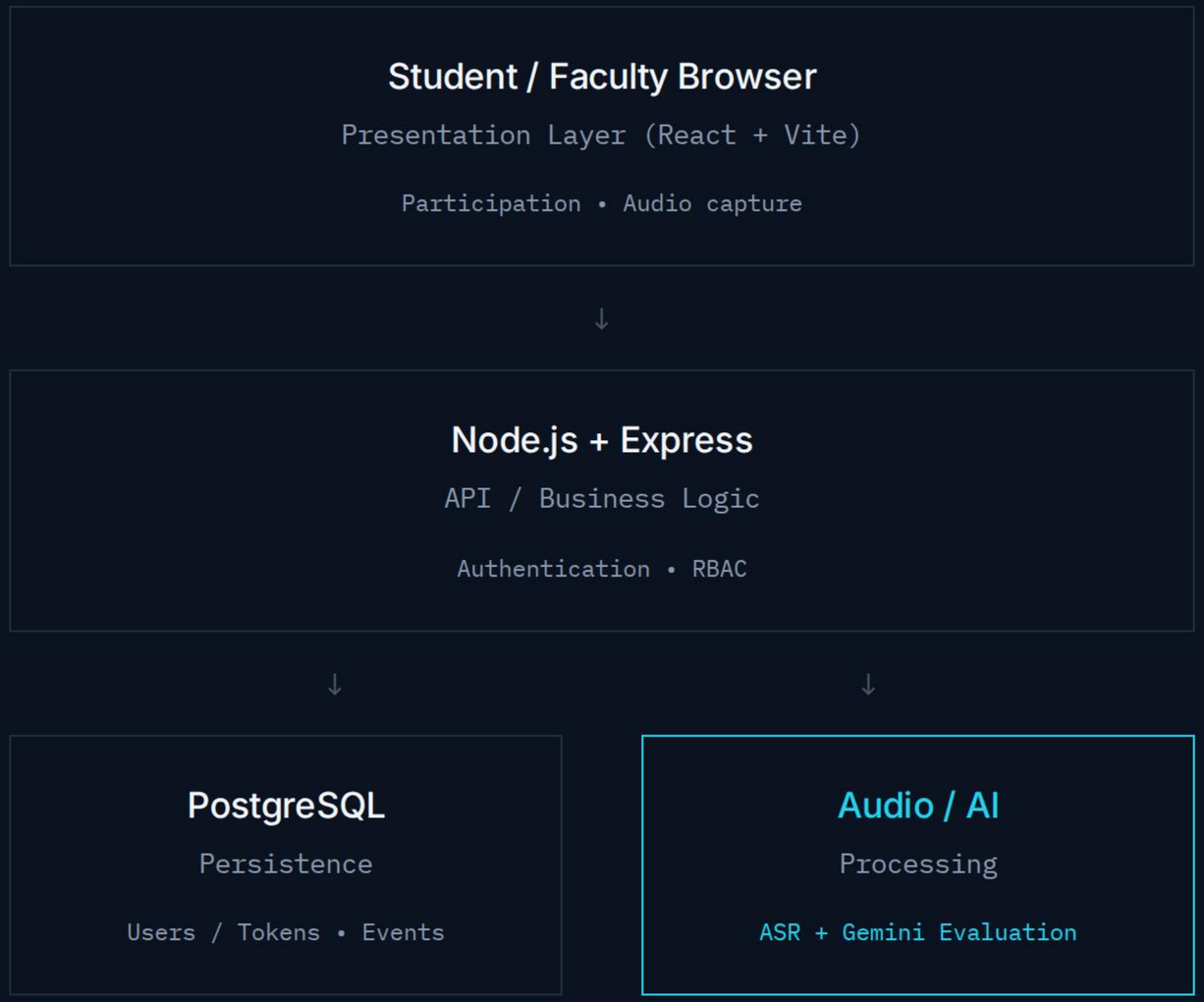
Automated token allocation can misinterpret responses or context.

DESIGN OBJECTIVE



CORE PRINCIPLE: AI ASSISTS THE EVALUATION — FACULTY RETAINS FINAL AUTHORITY.

System Architecture



Participation → Token Decision



AI-Assisted Evaluation — Human-Controlled Decision

INPUT

Teacher Context

- Topic
- Extra information

Student Response

- Recorded audio
- Transcript

AI EVALUATION

Analysis (Gemini)

- Relevance
- Correctness
- Reasoning

Output

Suggested token change

[−1 | 0 | +1]

HUMAN DECISION

Approve OR Override

→ Final Token Event

FINAL AUTHORITY = FACULTY

Security & Data Integrity

Identity

JWT + bcrypt

Access Control

Faculty / Student / Admin

Role-based authorization

Token Integrity

Running balance

Append-only event ledger

Reliability

Idempotency protection

Historical decisions preserved

Authentication

↓

Authorization

↓

Participation Event

↓

AI Suggestion

↓

Faculty Decision

↓

Immutable Token Event

THE FINAL TOKEN BALANCE IS DERIVED FROM RECORDED TOKEN EVENTS — NOT OVERWRITTEN HISTORY.

Current Prototype Status

CAPABILITY	STATUS
Authentication & RBAC	COMPLETE
Faculty Dashboard	COMPLETE
Raise-Hand Workflow	COMPLETE
Faculty Approval	COMPLETE
Real Browser Recording	COMPLETE
Audio Upload / Playback	COMPLETE

CAPABILITY	STATUS
Speech-to-Text	IMPLEMENTED
Teacher Context	COMPLETE
Gemini Evaluation	IMPLEMENTED
AI Token Suggestion	IMPLEMENTED
Faculty Approve/Override	COMPLETE
Token Balance & Ledger	COMPLETE

Student → Faculty → Recording → ASR → Gemini → Faculty → Token Ledger

CURRENT PROTOTYPE DEMONSTRATES AN END-TO-END WORKING PARTICIPATION-TO-TOKEN PIPELINE.

Roadmap → Production-Ready SmartToken

01

Offline Sync

Local actions → pending queue → connectivity restored → safe synchronization

02

Infrastructure

Object storage
Production deployment
Monitoring & backup

03

Assignments

Form submission → AI analysis → faculty approval

04

Rule Config

Activity-specific rules
Section configuration
Versioned history

05

AI Improvements

Better criteria
Teacher expectations
Confidence flags

TARGET SYSTEM: RELIABLE + AUDITABLE + FACULTY-CONTROLLED + AI-ASSISTED

AI IMPROVES THE EVALUATION WORKFLOW WITHOUT REPLACING ACADEMIC JUDGMENT.